WHITE PAPER · HWACHEON HI-ECO 21 HS × OIL & GAS

Hwacheon Hi-Eco 21 HS for Oil & Gas

High-speed CNC lathe for small-part precision turning where accuracy and cycle time matter most. This paper covers what the machine does, the oil & gas parts we produce with it, the alloys we run, the tolerances and finishes oil & gas buyers expect, and the documentation package.

8" OD × 12" L max turning ENVELOPE	**±0.0005" routine on critical features** TOLERANCE	**1994** SINCE
Same-week / rig-down direct RESPONSE		

Bearkat Blur · 8" OD × 12" L max turning · high-speed precision CNC turning center

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Executive summary

Machine: Hwacheon Hi-Eco 21 HS (shop nickname: *Bearkat Blur*). high-speed precision CNC turning center at B&R Productions, running from our New Waverly, Texas shop.

Application: High-speed CNC lathe for small-part precision turning where accuracy and cycle time matter most for Oil & Gas customers — Frac fleet operators, wellhead crews, wireline and coil-tubing service companies, downhole tool manufacturers, completion tool OEMs, subsea equipment vendors, produced-water handling equipment suppliers.

Envelope: 8" OD × 12" L max turning. **Tolerance:** ±0.0005" routine on critical features. **Traceability:** Material by heat and lot on every job.

The machine, in context

Bearkat Blur exists because certain oilfield and aerospace parts are small, high-quantity, and can't tolerate the setup time of a larger machine. Bushings, couplings, precision shafts, valve inserts, and any repeatable geometry in the 8-inch envelope — this machine cycles fast without giving up accuracy. When cycle time drives the quote, this is the machine that wins.

Why Bearkat Blur for Oil & Gas work

Not all oilfield work is large. Seat rings, valve inserts, bushings, couplings, and small precision shafts get ordered in quantity, and on those the quote is won on price per piece. 8" × 12", 4,000 RPM, 1,181 IPM rapids and a 12-station turret is a configuration built entirely around cycle time — without giving up the ±0.0005" the sealing features need.

What this looks like on a real part

A batch of valve seat inserts in Inconel 718 is a tool-wear problem before it's anything else. Each part is small, so a single insert edge does proportionally more of it, and 718 work-hardens the instant the edge stops cutting cleanly. Run the insert to a catalogue interval and the last parts in the batch have a different finish from the first. The interval gets set from measured wear on the opening pieces instead, and sealing surfaces get a fresh edge regardless of where that lands.

> **When this is the wrong machine**
>
> A hard 8-inch envelope. A 9-inch part goes to a Hi-Eco 35 cell, and forged stock needing heavy roughing goes to Hammer of Forge — 15 HP continuous is sized for finish-critical small work, not for taking scale off a forging.

Full specifications

Model	Hwacheon Hi-Eco 21 HS ("Bearkat Blur")
Type	High-speed CNC turning center — small-part specialist
Max turning capacity	8" OD × 12" L (B&R working spec)
Platform max swing	22.8" over bed; 11.02" turning diameter platform-max
Distance between centers	15.74" platform
Bar capacity	2" (through-spindle)
Chuck size	8"
Spindle nose	A2-6; 2.4" spindle bore
Spindle speed	4,000 RPM max
Spindle power	15 HP continuous / 20 HP 30-min rating
Turret	12-station
Rapid traverse	1,181 IPM on both X and Z axes
Control	Fanuc 18-T
Machine weight	~8,900 lb
Tolerance capability	±0.0005" routine on critical features
Materials	Stainless, brass, tool steel, Inconel 718/625, 17-4 PH, Monel

Values marked "platform" reflect the machine manufacturer's published spec range for this platform family; B&R's working spec is what we quote on this specific unit.

Who this serves in Oil & Gas

Typical buyer: Frac fleet operators, wellhead crews, wireline and coil-tubing service companies, downhole tool manufacturers, completion tool OEMs, subsea equipment vendors, produced-water handling equipment suppliers.

What's hard about oil & gas work: Chloride stress-corrosion cracking, sour service (H2S), high-cycle pressure loading, exotic superalloys, and rig-down clocks that turn every hour into money. Documentation trails required for API-adjacent work.

Typical Oil & Gas parts we produce on Bearkat Blur

- Small valve inserts, valve seats, retainer bushings
- Precision shafts and couplings under 8" OD
- Valve packing bushings and stem sleeves
- Wellhead adapter internals (small)
- High-quantity oilfield precision small parts

Above list is representative; if your part isn't shown, call and ask — most oilfield / aerospace / defense / industrial turned or milled work fits somewhere in the shop.

Alloys we run for Oil & Gas

Standard range: Inconel 718 (aged and annealed) · Inconel 625 · Super Duplex 2507 (UNS S32750) · Duplex 2205 · 17-4 PH (all conditions H900 to H1150) · Monel K-500 · Nitronic 50/60 · F22 · F91 · 4130/4140.

Alloy behavior is different in oilfield service. Inconel 718 in the aged condition (H900, roughly 40-45 HRC) is a different machining problem than annealed 718 — feeds, speeds, and tool geometry all shift. Super Duplex 2507 machined without cutting-temperature control develops brittle intermetallics that destroy the chloride-resistance the alloy was chosen for; a shop that shrugs when you ask about phase balance is a shop that will quietly ruin your part. 17-4 PH in Condition A machines like a well-behaved stainless; in H900 it's a different animal. Every one of these alloys we run weekly, not experimentally.

> **What "we run these weekly" means:** we stock or reliably source the common alloys in the industry's typical spec range. Sharp tooling, proven feeds and speeds, and process control specific to each alloy family. Your part is not the one we're experimenting on.

How we machine it — our 5-step process

This is the process B&R runs on every job, oil & gas or otherwise. The specifics of feeds, speeds, and tooling shift by alloy and machine; the discipline is universal.

1. **Material verification.** Every job starts with material traceability. Heat and lot numbers documented against the mill test report. For customer-supplied material, we verify against the CofC before touching the stock.

2. **First-article layout.** Stock is measured and laid out before any cutting starts. Concentricity, roundness, and squareness of the starting material established; if the stock is out of tolerance we call the customer before wasting time on a bad blank.

3. **Roughing with process control for the alloy.** Feeds, speeds, and coolant strategy set for the specific alloy family — sharp coated carbide (or PCD for high-volume Ti Grade 5), positive rake, high-pressure through-tool coolant on 2507 and Inconel to control cutting temperature. Interrupted cuts get extra attention because they cycle heat into the workpiece.

4. **Finish pass to spec.** Finish pass with a fresh insert, controlled DOC, and measured surface-finish targets. Critical features (bore concentricity, seat grooves, sealing surfaces) are held tighter than most job-shop lathes can offer.

5. **CMM verification + documentation.** Every critical dimension CMM-verified. First-article report generated. Certificate of conformance issued with delivery. If your quality group needs a documentation package beyond that, tell us with the RFQ and we'll be straight about what we can and can't produce in-house.

Tolerance and finish expectations in Oil & Gas

±0.0005" routine on critical sealing and mating surfaces. Bore concentricity and roundness held tighter than most job-shop lathes can offer. Surface-finish targets measured with a profilometer, not estimated. For seat grooves and packing bores, finish is often the difference between a sealing part and a leak-path.

Documentation and quality workflow

Material traceability by heat and lot on every job — non-negotiable for API-adjacent work. First-article layout and CMM verification standard. Certificate of conformance issued with delivery. If your quality group needs a documentation package beyond that, send us the requirement with the RFQ and we'll tell you straight whether we can meet it in-house or whether you need a shop with a formal QMS registration.

Response time and emergency work

Rig-down and fleet-down work prioritized over routine production. Same-day and next-day realistic when material is on the shelf. Call (936) 291-7827 direct — say "rig-down" and the phone gets to the shop floor. For emergency work off-hours, leave a voicemail; we route emergencies.

How we compare

Compared to buying finished components from an OEM: we're faster for one-offs and low-volume runs, and we hold traceability the OEM often won't share when you're outside their warranty window. Compared to a commodity job shop: we run the alloys weekly and won't ruin your metallurgy on cutting-temperature mistakes.

Working with B&R Productions

New customer or existing, the process starts the same way: call (936) 291-7827 or send a print through the quote form. Prints are accepted as PDF, STEP, IGES, DWG, or DXF; if you only have a sample part, we reverse-engineer routinely. NDAs on request, and prints stay in-shop rather than going out to third-party vendors. Note for defense and aerospace buyers: B&R is not ITAR- or AS9100-registered, so export-controlled prints and AS9100 flow-down programs need a registered shop — we'll tell you that on the first call, not after the PO.

Quotes typically come back within 24 hours for standard work, same-day for repeat customers on stocked materials, and near-immediately by phone for emergency and rig-down situations. Lead times are quoted honestly — we don't promise Wednesday and deliver next month.

Frequently asked questions

8 questions covering the Hwacheon Hi-Eco 21 HS platform and Oil & Gas work. Also indexed as FAQ schema for AI answer engines.

What's the max turning diameter on the Hi-Eco 21 HS?

8" OD × 12" L is B&R's working spec. The platform itself has a 22.8" swing over bed and can accept turning diameter up to 11.02" with 15.74" between centers. Bar capacity: 2" through spindle.

How fast is this machine — what's the spindle RPM?

4,000 RPM max spindle speed. Rapid traverse 1,181 IPM on both X and Z. Combined with the 8" chuck this is our high-cycle-count small-part specialist.

What's the turret capacity?

12-station turret. Enough tool positions for most single-setup precision turning jobs including grooving, threading, and finish operations.

What kind of oil & gas customers does B&R Productions work with?

Frac fleet operators, wellhead crews, wireline and coil-tubing service companies, downhole tool manufacturers, completion tool OEMs, and subsea equipment vendors. We work with new customers regularly — first-time orders don't get worse service.

Do you handle rig-down and emergency work?

Yes. Same-day and next-day are realistic when the material is on the shelf (Inconel 718, 625, Super Duplex 2507, 17-4 PH, 4140/4340). Call (936) 291-7827 direct and say "rig-down" — the phone gets to the shop floor, not a form.

Do you machine to API 6A / API 6D tolerances?

Yes — API-adjacent tolerances are routine, and material traceability by heat and lot is standard on every job. Worth being precise about the wording, though: we machine to the tolerances and finishes those specs call for, but B&R is not an API-licensed facility. If your buyer requires parts from an API-licensed shop with a monogram, that's a different vendor. If they require parts machined to print at API-class tolerances with full traceability, that's us.

How do you keep Super Duplex 2507 from losing its chloride resistance during machining?

Controlled cutting temperature — moderate SFM, aggressive feed with sharp coated carbide, high-pressure through-tool coolant. The failure mode nobody wants (sigma-phase intermetallics forming from overheating) shows up months later in service. We run 2507 weekly — this is a process we've internalized, not a spec we're learning on your part.

Can you reverse-engineer discontinued OEM oilfield parts?

Yes — routine work. Send a worn sample and any print or spec you have. We produce the replacement with material traceability the OEM often won't share when you're outside their warranty window.

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Ready to talk about your Oil & Gas work?

Send us a print or a sample photo and specs. Emergencies: call direct and say "rig-down."

Published by B&R Productions — a precision CNC machining shop in New Waverly, Texas, in business since 1994. Serving oil & gas, aerospace, defense, and industrial customers across Texas and the Gulf Coast.

Written by the B&R Productions team. Machine specs verified against manufacturer data. Alloy and process notes reflect shop practice as of 2026-01-15.